One continuation-connected family and two bracketed Floquet events in the Li–Li–Liao unequal-mass catalog

(claim-gated letter; scientific status OPEN)

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Abstract

Li, Li and Liao report 135,445 unequal-mass non-hierarchical planar periodic orbits of the free-group word $bABabaBAba$ on a 0.001 mass grid over $m_1 \in [0.8, 1.1]$, $m_2 \in [0.7, 1.2]$ at $m_3 = 1$, with 13,315 linearly stable samples and 620 adjacent stable/unstable cells [2]; a 2023 analysis of the same table reads it as two independent sets in a corrected scale-invariant period/angular-momentum projection [3]. Neither the catalog nor its stability labels is a result of this work. We treat family identity as branch-preserving continuation connectivity rather than as the appearance of a projection, and the linear-stability boundary as the preimage, under the sheet map $\Phi : (m_1, m_2) \mapsto (\alpha, \beta)$, of the boundary of the universal spectral domain of the reduced trace polynomial — the walls $G_+ = 0$, $G_- = 0$, $\Delta = 0$ and the exact vertices $(0, -4)$, $(4, 4)$, $(8, 28)$ — rather than as a rectangular S/U mask. That algebra is classical; only its preimage on this catalog sheet is computed here. At the rungs the artifacts themselves declare: the sampled set is *supported as* one continuation-connected component, on 26 adversarially chosen links crossed bidirectionally without loosening a gate, with chart independence shown on exactly one of the 26; the corrected (T_{si}, L_{si}) projection on this sheet is folded and non-injective, which defeats the inference from two visible branches to two dynamical families without disproving the 2023 analysis on its own terms. Two events are *independently reproduced* as bracketed physical mechanisms in a second arbitrary-precision program: a generic physical $+1$ crossing bracketed in m_2 to width 1.578×10^{-9}, and an opposite-Krein Hamiltonian–Hopf collision bracketed to 6.268×10^{-9}. One rung below, three mixed roots of $G_+ = G_- = 0$ — preimages of the $(4, 4)$ vertex — are *high-precision supported* only: localized by Newton residual, not by a sign change and not by interval arithmetic. All 620 transition cells are localized once each and assigned to one of seven mechanism polylines. Not earned: no theorem of connectivity, no interval-arithmetic existence proof, no transversality or nondegeneracy certificate, no complete critical set and no complete critical graph. No fourth organizer, no fold and no daughter classification is claimed. The release flag is `false` and the scientific status is OPEN. Throughout, "stable" means planar linear Floquet stability only — not nonlinear, not KAM, not spatial — and no priority claim is made.

1 The question, and what it is not

The question is finite and it is posed inside somebody else's table. Fix the word $bABabaBAba$, fix $m_3 = 1$, and fix the mass rectangle $m_1 \in [0.8, 1.1]$, $m_2 \in [0.7, 1.2]$ of area 0.15 on which Li, Li and Liao sampled 135,445 orbits at grid step 0.001, labelling 13,315 linearly stable and leaving 620 adjacent cells across which that label flips [2]. Two questions about that frozen object are then answerable rather than philosophical. First: is the sampled set one family or several, where *family* means branch-preserving continuation of the periodic orbit inside the rectangle and not the appearance of a projection — the distinction that separates this reading from the two-set reading of the same table [3]. Second: what is the stability boundary inside the rectangle, resolved by mechanism, once one stops treating stability as a binary mask and computes instead the preimage under Φ of the boundary of the universal spectral domain of the reduced monodromy.

What this is not. It is not the general three-body problem and settles nothing about it. It is not a new catalog: the orbits, the stable labels and the broad stable region belong to [2] and enter here as fixed input, pinned by the content digest of the published supplementary table. It is not a theorem: connectivity is asserted only as a bounded numerical continuation statement inside the declared rectangle and normalization, and a chart singularity is not a moduli-space disconnection. It is not a completeness result for the critical set; a finite search at step 0.001 cannot see two roots inside one cell, an even number of crossings, or a tangency. And it is not a statement about the stability dynamicists usually mean: "stable" here denotes spectral stability of the linearised return map for planar motion — not nonlinear or Lyapunov stability, not KAM, not stability against non-planar perturbation. A spectrally stable Hamiltonian orbit can be nonlinearly unstable through resonance, and nothing below bears on that.

Rungs. Every statement carries the rung of the artifact that produced it, and the rungs are not interchangeable. **Screening** is IEEE binary64 shooting, Newton correction and event localization at two frozen gates — maximum absolute event 2×10^{-8}, maximum periodic closure 10^{-7} — never loosened anywhere in this work; a float64 point is a proposal. **High-precision supported** means recomputed at 60 digits in an independent Julia BigFloat program with canonical Jacobi monodromy and the physical E^ω/E symplectic quotient, but localized by Newton residual rather than by a bracket. **Independently reproduced** means the event is localized by a sign change of the event function in a program sharing no dynamics, shooting or Floquet implementation with the screening stack, with closure, symplectic-defect, reciprocal-pairing, leakage and Krein diagnostics recorded on both flanks. One rung of separation between the bracketed events and the mixed roots is not a stylistic hedge: it is the difference between a sign change and a residual.

2 The universal spectral frame

For the centre-of-mass-reduced planar problem the reduced monodromy is symplectic on the physical E^ω/E quotient, and its characteristic polynomial is determined by two invariants (α, β). Spectral stability of the reduced return map is then a region in the (α, β) plane bounded by three curves: the walls $G_+ = 0$ and $G_- = 0$ where an eigenvalue reaches $+1$ or -1, and the discriminant parabola $\Delta = 0$ where two unit-circle pairs collide. The three curves meet at the exact vertices $(0, -4)$, $(4, 4)$ and $(8, 28)$. This algebra is classical symplectic Floquet and Krein theory [4, 1] and is not a contribution of this work. What is computed here is the preimage $\Phi^{-1}(\partial\mathcal{D})$ on one catalog sheet, where $\Phi : (m_1, m_2) \mapsto (\alpha, \beta)$. The gain is that a stability transition acquires a mechanism: a G_+ crossing, a G_- crossing, or a Δ collision, with the $(4, 4)$ vertex as the organizing preimage where a G_+ and a G_- wall meet. The free-group word is carried as catalog metadata only; the artifact records that topology is not used as family identity.

3 One continuation-connected component

Within the frozen catalog and the declared normalization the sampled set is *supported as* one continuation-connected component. The evidence is adversarial rather than exhaustive. A spanning tree on the 135,444 mass-adjacent edges was built, and 26 links were chosen to be as hostile to connectivity as the frozen data permits: five macroscopic bottleneck cuts of that tree, the twenty globally worst chart jumps, and one deliberately pathological invariant-near-duplicate bridge whose endpoints lie 0.0617 apart in mass — sixty-one grid steps — and which is therefore not a tree edge at all. Each of the 26 was crossed bidirectionally without loosening a gate. Chart independence was demonstrated on exactly one of the 26, the pathological bridge, by re-crossing it in a generic strict-periodic chart.

The two-set reading is addressed on the mechanism, not on the labels. On this sheet the corrected scale-invariant (T_{si}, L_{si}) projection is folded and non-injective: 590 mass-adjacent edges carry a determinant sign change and assemble into one fold locus, and 86 far-separated mass pairs agree to standardized invariant distance below 10^{-4}. A folded non-injective projection can show two visible branches for one connected sheet, so the inference from two branches to two dynamical families does not follow. This does not disprove the 2023 analysis on its own (T, L) terms [3], and no claim is made that it does.

Three caveats travel with this result and are not negotiable. The certificate covers 26 links, not 135,444. Chart independence rests on one of them. And separation below the published 0.001 mass grid cannot be excluded by any of it.

4 Two bracketed events and three mixed roots

Two critical events are independently reproduced as bracketed physical mechanisms. On the slice $m_1 = 0.8023446113666945$ the lower transition is a generic physical $+1$ crossing of the reduced return map, bracketed in m_2 to width $1.5776460402494633 \times 10^{-9}$. On $m_1 = 0.8022889780964406$ the upper transition is an opposite-Krein Hamiltonian–Hopf collision, bracketed to width $6.2677772144967 \times 10^{-9}$, with two upper-half unit modes of opposite Krein sign on the stable flank and none on the unstable flank — the Krein signature, not merely the eigenvalue positions, is what identifies the mechanism.

One rung below, three mixed roots of $G_+ = G_- = 0$, the preimages of the $(4, 4)$ vertex, sit at $(0.9292391921, 0.8853664625)$, $(0.9967681995, 0.9560193602)$ and $(1.0495531867, 1.1294758073)$. Each is recomputed at 60 digits in the independent program, with relative event norm of order 10^{-14}, but each is localized by Newton residual rather than by a sign-change bracket. Their own artifacts declare this rung, and the graph's node labels are more generous than the artifacts warrant; the artifacts govern. Two independent BigFloat programs agree on all five objects to between 8.9×10^{-27} and 5.7×10^{-25} in mass.

5 The 620-cell decomposition, and what it rests on

All 620 published transition cells are localized once each — no cell is localized twice and none is left as a Newton failure — and each is assigned to exactly one of seven mechanism polylines: 198 cells on G_+ crossings, 168 on G_- crossings and 254 on Δ collisions. Six further label-invisible event-sign polylines from the full-domain sweep carry 153 cells in components of 19, 24, 29, 23, 34 and 24, giving the thirteen-edge assembly at HEAD.

Object	Location	Localization	Rung
Lower +1 crossing	$m_1 = 0.80234461$	m_2 bracket 1.578×10^{-9}	independently reproduced
Upper H–Hopf collision	$m_1 = 0.80228898$	m_2 bracket 6.268×10^{-9}	independently reproduced
Mixed root, principal left	$(0.92923919, 0.88536646)$	Newton residual $\sim 10^{-14}$	high-precision supported
Mixed root, secondary left	$(0.99676820, 0.95601936)$	Newton residual $\sim 10^{-14}$	high-precision supported
Mixed root, principal right	$(1.04955319, 1.12947581)$	Newton residual $\sim 10^{-14}$	high-precision supported

Table 1: The five objects, each at the rung its own artifact declares. A 60-digit bracket is a bracket, not an interval-arithmetic existence proof; a Newton residual is weaker still.

The honest reading of that assembly is a mixture, not a certification. The artifact never uses the word "certified" for an edge. By the edges' own `estimators` fields, seven of the thirteen polylines rest on float64 screening alone and six carry independent BigFloat corroboration; of the 620 localized cells, 462 are float64 and 158 BigFloat. Under this project's own ladder, float64 output stops at screening, so more than half the assembled decomposition is a proposal rather than a verified object. The artifact describes itself as a partial graph with `release_ready: false`.

6 What is not earned

The critical graph is not finished. Of the 26 edge endpoints, 24 are attached: two finite-lattice sweep ends remain unclassified, at $(0.892, 0.7530796376143668)$ and $(1.042, 0.8579752021443232)$. The edge incidence splits into two components — the trace-collision curve is still its own component — and three declared nodes carry no edge incidence at all. The assembler's incidence summary is diagnostic and feeds no release gate, so graph connectivity is reported here as an open structural fact rather than as a gate that has been failed.

The event gate is finer than the reproducibility of the quantity it gates. Across the 775 localized roots in the residual-margin audit the worst absolute event is $1.989798081858396 \times 10^{-8}$ against the frozen gate of 2×10^{-8}: 99.4899% of the budget, cleared by 0.5101%, with 222 roots above 10^{-8} and 12 above 95% of the gate. That figure is a property of one chart, one tolerance and one machine rather than of the roots. Re-deriving the same monodromy through a rotation at identical tolerance moves the event past the gate on 345 of the 620 cells and past ten times the gate on 144, concentrated in G_+ (126/198) and Δ (217/254); G_- (2/168) is the only mechanism whose gate compliance survives that re-derivation. Conditioning is recorded for none of the 775 roots. The closure certificates are also optimistic: they are written from the corrector's screening integration at relative tolerance 2×10^{-10} while the event is evaluated at 5×10^{-13}, and on a seeded sample of 160 roots the recorded closure is optimistic by a median factor 48.6, with four of the 160 exceeding the frozen 10^{-7} closure gate on re-measurement, the worst at 1.543×10^{-7}. Mechanism labels are not immune either: BigFloat escalation of sixteen float64-failed canary cells returned four of them as Δ collisions where float64 had recorded G_+.

Three classifications are explicitly not claimed. The secondary-left birth is not a certified nondegenerate fold: transversality is real, with $\partial G/\partial m_1 = 28.045563274511$ converging under refinement, but the second-derivative stationarity test does not converge to zero — it converges to -30 — so the fold normal form is not established. The secondary-right terminus carries physical evidence only and is typed as an endpoint, not an organizer; no fourth mixed organizer is claimed anywhere in this letter. The lower +1 daughter is not classified: the classifier can emit only one outcome by construction, so its recorded class is an assertion of the script rather than a discrimination among the alternatives, and reconnection is excluded only over an m_2 window of 5.60×10^{-5}.

Sign-topology coverage is withdrawn. The coverage flag was withdrawn after it was found true only on unmeasured territory, and it remains withdrawn. The densest full-domain audit at HEAD probes 121 scan lines at m_1 step 0.0025 across the declared m_2 range with an allowed gap of 0.0009: of 273 planned probes, 229 converged and 44 failed, and 25 curve endpoints remain unrefined. The deficiency is probe convergence near the mass-domain faces, not sampling density alone. The BigFloat verifier compounds this by not enforcing the frozen event gate on arm extensions: two recorded successes carry absolute events of 1.15×10^{-7} and 2.27×10^{-7}, an order of magnitude past the gate.

7 Completeness, in the bounded sense

Completeness is frozen, and bounded, and the bound is small. Searched as a two-dimensional region, the frozen neck raster covers $m_1 \in [0.997, 0.999] \times m_2 \in [0.993, 1.012]$ at step 10^{-4} with 4011 samples, all 21 lines separated and a minimum resolved unstable gap of 3.0×10^{-4}: 0.0253% of the declared area. The only other completeness input is a twelve-proposal off-grid active-learning screen in one pocket, $m_1 \in [0.80005, 0.80276] \times m_2 \in [0.75266, 0.75407]$, 0.002539% of the area, and its own artifact describes it as "AI proposals plus float64 screening only; not scientific discovery evidence". The two together are 0.0279% of the declared domain. Everything else is probed on a measure-zero union of one-dimensional scan lines.

So "completeness" here can honestly mean only this: no additional critical object was detected by a finite sign-change sampling at those resolutions. The scope registry records that even-order root pairs and tangencies are explicitly not excluded, and sub-resolution loops and curves lying between scan lines are not excluded either. Under this project's own invalidation rule the later full-domain sweep does not invalidate the frozen certificate — neither certificate source changed bytes — but neither does the certificate lend any support to completeness of the critical *set*: the sweep reports gate-passing roots that match no committed edge, and components absent from the thirteen-edge graph. Whether the interior is clean is therefore recorded here as unresolved.

8 Reproducibility, and status

The upstream supplementary table is downloaded at run time and rejected unless its Git blob is exactly `79b2963df43e62201c35690bfc22bec166132427`. Python and Julia environments are pinned by lockfile and manifest. The release bit is not writable by hand: only the assembler may set `release_ready`, the manuscript's generated sections are emitted from frozen claim records, and a release build refuses unless the status is `solved`. Re-running the assembler at HEAD reproduces the committed critical-graph artifact byte for byte and exits nonzero, which is the intended behaviour while the problem is open.

Two of the assembler's fourteen release conjuncts are false at HEAD: no unclassified edge endpoints, and sign-topology coverage. The novelty freeze is separately pending. Accordingly `release_ready` is `false`, the scientific status is OPEN, and this letter reports what is earned rather than a solution. No priority claim is made.

References

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[4] V. A. Yakubovich and V. M. Starzhinskii. *Linear Differential Equations with Periodic Coefficients*. Wiley, New York, 1975. Translated from the Russian; standard reference for the strong-stability and Krein-signature theory of symplectic monodromy used here as a frame.